

Engineering Design & Tech Stack

ECITY — Mobile Shop Management Web App

Version 1.4 · 2 September 2026

TypeScript stack, architecture, self-hosted on one server

Companion documents: PRD, Module Breakdown, Deployment Guide

1. Purpose and Constraints

This document specifies **how** to build the product described in the PRD, for a small team — realistically one developer — whose strongest language is **TypeScript / JavaScript**, with **shadcn/ui** for the interface, and with a hard eye on **monthly running cost**.

Constraints that shaped every choice below:

Constraint	Consequence
TypeScript/JavaScript is the familiar language	One language across frontend, backend, scripts, tests and migrations. No Java, Go, Python or PHP in the critical path
One developer, ~30 modules of scope	Prefer a single deployable application over microservices; prefer boring, well-documented tools over clever ones
Money, stock and IMEI records must be exactly right	A relational database with real transactions and constraints. Not negotiable
Reporting and analytics are half the product	SQL-first data layer. The reports are joins and aggregates, and a document database would make them painful
Monthly cost must stay small and predictable	One rented Linux server running everything in Docker ; no managed-service subscriptions, no per-seat SaaS in the critical path. Target under ₹1,000/month
shadcn/ui requested for styling and components	React + Tailwind CSS, which fixes the frontend framework family

2. Recommended Stack

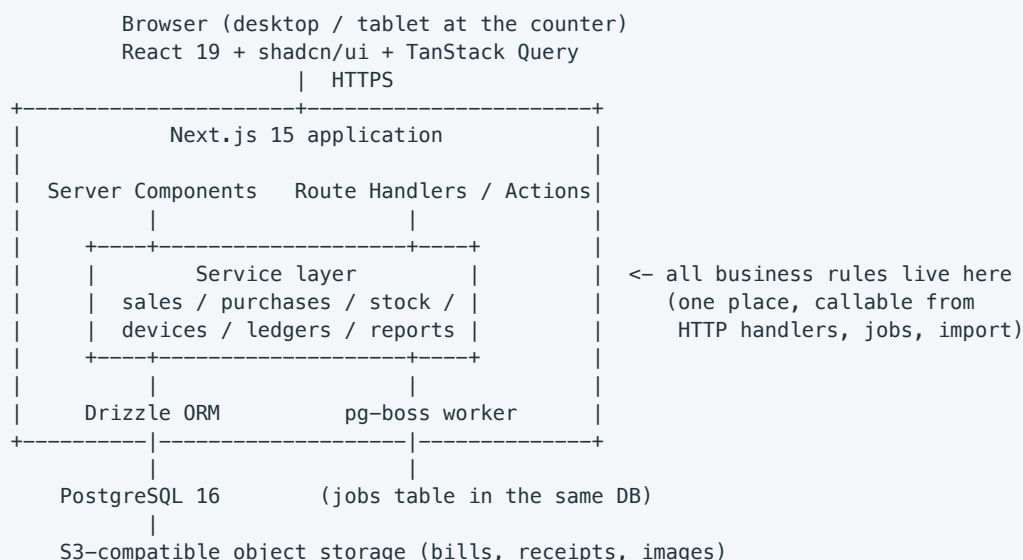
Layer	Choice	Why this one
Language	TypeScript 5.x (strict)	Your language; one type system from the database row to the React prop. Strict mode is what makes it worth having
Framework	Next.js 15 (App Router)	Frontend and backend in one deployable TypeScript project. Server Components keep heavy report queries on the server; Route Handlers give you a normal REST API for the counter screens
UI components	shadcn/ui + Radix UI	As requested. Components are copied into your repo, so you own and can adapt them — right for a dense back-office app with tables, dialogs, comboboxes and command palettes
Styling	Tailwind CSS v4	Required by shadcn/ui; keeps a 40-screen app visually consistent without a growing stylesheet
Forms & validation	React Hook Form + Zod	One Zod schema validates the browser form and the server handler, and infers the TypeScript type. Removes a whole class of bugs in a form-heavy product
Server data / cache	TanStack Query	Caching, background refetch, optimistic updates and retry — matters on the billing screen and on a shaky shop connection
Tables	TanStack Table (headless, styled with shadcn)	Inventory, sales and report grids need sorting, filtering, pagination and column visibility. Do not hand-roll this
Charts	Recharts	React-native API, adequate for the dashboards and analytics in PRD §6.15
Database	PostgreSQL 16+	Transactions, foreign keys, check constraints, partial and composite indexes, window functions for analytics, full-text search for global search, numeric/bigint for money. Everything this product needs is in the box
ORM / query layer	Drizzle ORM	TypeScript-first, generates SQL you can read, migrations are plain SQL files, and dropping to raw SQL for a report is trivial. <i>Alternative:</i> Prisma, if you prefer its DX — both are fine; Drizzle wins on report-heavy work and cold-start cost
Auth	Own session layer — random token in an httpOnly cookie, SHA-256 hash stored in Postgres	Sessions in your own database, revocable, with sliding idle expiry. Auth.js v5 was the original choice, but its Credentials provider only supports JWT sessions — database sessions are not available with it, and PRD FR-1.1 requires revocable server-side sessions. Roughly 150 lines (src/server/auth/session.ts), no vendor, no library to fight
Authorisation	Own permission layer (requirePermission(user, perm, branchId))	Roles are editable sets of permissions plus a branch scope; no library models this well. Optionally reinforced with Postgres Row-Level Security later
File storage	Cloudflare R2 (S3-compatible)	Bill photos, receipts, product images. Free tier covers 10 GB with zero egress fees. Keep it behind a small internal interface so any S3-compatible provider can replace it

Layer	Choice	Why this one
Background jobs	pg-boss (job queue inside Postgres)	Alerts, nightly rollups, import processing. Runs as a second container beside the app — it needs a long-lived process, which is one more reason not to deploy serverless. No Redis, no extra service, no extra bill
PDF / print	Browser print + CSS @page for A4 and 80 mm; @react-pdf/renderer for downloadable invoice PDFs	Avoids running headless Chrome, which is the expensive way to make a PDF
Excel / CSV export	exceljs , streaming CSV	Handles the export requirements in FR-25.4 and FR-33.3 without loading everything into memory
Testing	Vitest (unit), Playwright (end-to-end)	One test runner for TS; Playwright to prove the six core business flows in PRD §7 keep working
Error tracking	Sentry	Free tier is sufficient at this scale
Hosting	One Hetzner CX22 VPS , everything in Docker	App, PostgreSQL, pg-boss worker and Caddy (TLS) as four containers on one machine. ≈ ₹550–750/month all in. See §6 and the Deployment Guide
CI/CD	GitHub Actions	Typecheck, lint, test, build an image, push to GHCR, SSH to the server and restart. The server never compiles anything

2.1 Deliberate rejections

Rejected	Reason
MongoDB / Firebase / any document store	The product is inherently relational — sales, items, devices, ledgers, branches — and needs multi-row transactions and aggregate reporting. This choice would be paid for daily
Serverless hosting (Vercel, Lambda) for this app	Three costs with no matching benefit here: cold starts of 300 ms–1 s on the billing screen, a connection storm against Postgres that needs a pooler to survive, and no long-lived process for the pg-boss worker. A single always-on Node process has none of these problems and is cheaper
Managed database as a service (Supabase, Neon, RDS)	₹1,300–2,200/month for something a Docker container does for ₹0 on a box you are already renting. What you actually buy is managed backups — worth it at scale, not at one shop. §6 and the Deployment Guide replace it with three backup layers
Separate NestJS/Express backend repo	Doubles the deployment, auth plumbing and type-sharing work for a solo developer with no benefit at this scale. Keep one app; the service layer inside it is the seam if you ever split
Microservices	There is no scaling or team problem here that they solve, and several correctness problems they create
Floating-point money (float , double , JS number for amounts)	Rounding errors in a system whose whole purpose is reconciling cash. Store integer paise in bigint
A BaaS for auth (Clerk, Auth0) at \$25+/mo	Cost with no benefit: your roles and branch scoping are custom regardless
Prisma Accelerate / paid data proxies	Not needed at this connection count

3. Architecture



One rule keeps this maintainable: HTTP handlers, background jobs and the CSV importer all call the *same* service functions. `sellDevice()` is written once, so a device sold through the UI, through an import or through a correction job produces the same stock movement, the same ledger rows and the same `device_event`.

3.1 Repository layout

```

/app                Next.js routes (route groups per module)
/(auth)            login, reset
/(app)/[branch]    dashboard, billing, inventory, purchases, ...
/api               route handlers
/components         shadcn/ui primitives + app components
/server
  /db              drizzle schema, migrations, seed
/services          sales, purchases, stock, devices, ledger, reports, search
  /external-import adapters + mapping for the other billing system's Excel
                  (see the note below – only the adapter knows the layout)
/auth              session, permissions, branch scope
/jobs              pg-boss workers: alerts, rollups, imports, backups
/lib               zod schemas, money helpers, formatting, constants
/tests             vitest unit, playwright e2e

```

4. Data Layer Decisions

These seven decisions are the difference between a system that reconciles and one that does not.

4.1 Money is bigint paisa. Every amount column is integer minor units. A `Money` helper type handles arithmetic, and formatting to ₹ happens only at the display edge. No `float`, no `number` arithmetic on amounts in JavaScript.

4.2 Balances are derived, never stored as mutable counters.

Customer outstanding, supplier outstanding, cash-drawer cash and account balances are all sums over append-only ledger tables. Cache them in a summary column if a report demands it, but the ledger is the truth and a nightly job must be able to prove the cached value equals the recomputed one.

4.3 `device_event` is append-only and is the source of the IMEI history.

```
create table device_event (
    id          bigint primary key,
    device_id   bigint not null references device unit(id),
```

```

seq          int      not null,          -- ordering within a device
event_type   text      not null,          -- purchased | received | transferred_out |
                                           -- transferred_in | reserved | sold | returned |
                                           -- inspected | reclassified | repaired |
                                           -- damaged | lost | adjusted | voided

occurred_at  timestampz not null,
branch_id    bigint references branch(id),
from_branch_id bigint references branch(id),
to_branch_id  bigint references branch(id),
ref_type     text,                        -- purchase|sale|transfer|return|adjustment
ref_id       bigint,
actor_id     bigint references app_user(id),
payload      jsonb  not null default '{}',
             unique (device_id, seq)
);
create index on device_event (device_id, seq);

```

No **UPDATE** and no **DELETE** are permitted on this table — revoke those privileges from the application role. The device history page in PRD §6.21 is one indexed read of this table joined out to its referenced documents.

4.4 A device's IMEIs are a separate table, not `imei_1` / `imei_2` columns. A handset can carry more than one IMEI, and some carry more than two. Storing them as numbered columns means a schema migration, a data backfill and a rewrite of every form, query, importer and report the day a third one appears. Storing them as rows costs one join today and nothing later.

```

create table device_identifier (
  id          bigserial primary key,
  device_id   bigint  not null references device_unit(id) on delete restrict,
  imei        text      not null,
  slot        smallint not null default 1,      -- 1 = first SIM slot
  is_primary  boolean  not null default false,
  created_at  timestampz not null default now(),
  constraint imei_digits check (imei ~ '^[0-9]{14,17}$'),
  unique (imei),                                -- unique across the whole business
  unique (device_id, slot)
);

-- exactly one primary identifier per device
create unique index device_identifier_one_primary
  on device_identifier (device_id) where is_primary;

-- partial-IMEI search
create index device_identifier_imei_trgm
  on device_identifier using gin (imei gin_trgm_ops);

```

`device_unit.primary_imei` is kept as a cached copy, maintained by a trigger on `device_identifier`, so that lists, invoices and exports do not join for the common case. It is a display convenience and never the source of truth — all matching, validation and search run against `device_identifier`.

The UI shows one IMEI field in v1, and that is a setting, not a limitation. `business_setting.imei_slots` (default 1) controls only how many IMEI inputs the purchase and product forms render. It has no effect on validation, storage, search, import or export: the service API always takes and returns `imeis: string[]`, and a device that arrives from an import with three identifiers is stored, searched and reported with all three while the setting still reads 1. Raising it to 2 or 3 later is a row in a settings table — no migration, no backfill, no release.

```

// service API – the shape never changes when the setting does
createDeviceUnit({ productId, branchId, mainType, isNewCut, imeis: string[], ... })
findDeviceByImei(imei)           // matches ANY identifier of any device

```

4.5 The GLOBAL / NEW CUT rule is enforced by the database, not only by code.

```

alter table device_unit add constraint new_cut_only_global
  check (is_new_cut = false or main_type = 'GLOBAL');

```

Application validation can be bypassed by an import, a script or a bug. A check constraint cannot.

4.6 Concurrency on device sale. Two tills must never sell the same IMEI. Guard it with a conditional update inside the sale transaction rather than an application-level read-then-write:

```
update device_unit
  set status = 'SOLD'
  where id = $1 and status = 'IN_STOCK' and current_branch_id = $2
  returning id; -- zero rows returned => abort the sale with a clear message
```

4.7 Indexes that must exist from day one. `device_identifier(imei)` unique plus its trigram index; `device_unit(current_branch_id, status, main_type)`; `device_event(device_id, seq)`; `sale(branch_id, created_at)`; `sale_item(device_id)`; `customer(phone)`, `supplier(phone)`; and GIN trigram indexes on customer, supplier and product names for global search. Add a `pg_trgm` extension for partial IMEI and name matching.

4.8 Reporting strategy. Live SQL for anything within the current month. For year-range analytics, a nightly `pg-boss` job writes per-branch, per-day rollups (sales, cost, profit, units, payment mix, stock value) and the analytics pages read rollups for closed days plus live queries for today. This keeps PRD \$9.1's five-second target reachable without a warehouse.

5. Cross-Cutting Implementation Notes

Authorisation. Every route handler and server action begins with `requirePermission(user, permission, branchId)`. Write an automated test that enumerates every endpoint and asserts a 403 for a user lacking the permission or the branch — this is the single most valuable test suite in the project, because the failure mode (Branch A staff reading Branch B money) is invisible until it is embarrassing.

Branch scope. Resolve the active branch (or the permitted branch set for consolidated views) once per request and thread it through the service layer as an explicit argument. Never read it from a global.

Idempotency. The billing screen generates a UUID per bill attempt and sends it as an idempotency key. The server stores it uniquely; a retry after a dropped connection returns the original sale instead of creating a second one. Do the same for payments and transfers.

Audit. A thin wrapper over the write path records actor, entity, action and changed fields into `audit_log` automatically, so a new feature is audited without the developer remembering.

Barcode / IMEI scanning. Scanners behave as keyboards. The billing search input listens for fast character bursts terminated by Enter and treats them as a scan; no drivers, no native app. A scan resolves through `device_identifier`, so scanning the label's second IMEI finds the device just as well as the first.

Printing. Two print stylesheets — A4 invoice and 80 mm thermal — driven by CSS `@page` and `@media print`. Downloadable PDFs use `@react-pdf/renderer` on the server. Avoid headless Chrome; it is the single most expensive thing you could add to your hosting bill.

Uploads. The browser requests a short-lived signed upload URL and posts directly to object storage; the server stores only the key. Downloads are served through short-lived signed URLs, never a public bucket.

Integrating the other billing system. The shop keeps a second system for NEW items and exports Excel from it daily (PRD \$6.22). The column layout is unknown at design time, so isolate it: `external-import/types.ts` defines a canonical row shape that the rest of the pipeline is written against, `adapters/*.adapter.ts` is the only code that knows the real column names, and `mapping/legacy.mapping.json` holds the column-name mapping so a changed header is a config edit rather than a release. Staging and preview before commit, idempotency on `(source, external invoice no, external line id)` plus a file hash, and apply through the `same` service functions as manual entry — never straight into tables. The one rule that makes this safe rather than merely tidy: a device whose `sales_channel` is `EXTERNAL` cannot be sold on the ECITY billing screen at all, so the two systems can never invoice the same IMEI.

Environments. `local` (Docker Compose on your laptop) → `staging` → `production`. Staging is either a second small Hetzner server or a second Docker stack on the same box with its own database and port; either way it costs little and it is what stops a bad migration reaching real sales data. Migrations run in CI against staging before they are allowed near production.

6. Hosting and Monthly Running Cost

Everything runs on **one rented Linux server**. The app, the database, the background worker and the TLS proxy are four Docker containers on the same machine, talking to each other over Docker's internal network. There is no managed database, no serverless platform and no per-seat subscription anywhere in the critical path.

Prices are indicative for **September 2026**, converted at approximately ₹88 = \$1. Confirm on the provider's own pricing page before committing — regional prices differ from headline list prices.

6.1 The recommended setup

Item	Choice	₹ / month
Server	Hetzner CX22 , Singapore — 2 vCPU, 4 GB RAM, 40 GB SSD	~₹400–600
Automated snapshots	Hetzner backups (+20% of server)	~₹90
Object storage + off-site backups	Cloudflare R2, free tier (10 GB, zero egress)	₹0
Transactional email	Resend free tier (3,000/month)	₹0
Error tracking	Sentry Developer, free	₹0
Domain	<code>.in</code> , ~₹800/year	~₹70
Total		≈ ₹550–750

Hetzner is an independent German hosting company — you buy directly at hetzner.com/cloud, not through AWS. Their nearest region to India is **Singapore** at roughly 60–90 ms, comfortably inside the 300 ms interaction target in PRD §9.1. They also bill from the EU without adding 18% GST, which widens the gap against Indian-region providers.

Step-by-step setup, hardening, deployment pipeline, backup scripts and the operational runbook are in the companion **Deployment Guide**.

6.2 If an India-region server is wanted

Provider	Region	2 vCPU / 4 GB, ₹ / month
DigitalOcean	Bangalore	≈ ₹1,060 + 18% GST
AWS Lightsail	Mumbai	≈ ₹1,050 + 18% GST
Linode / Akamai	Mumbai	≈ ₹1,060 + 18% GST

Same architecture, same containers, about ₹800 more per month, latency down to roughly 20 ms. Worth it only if counter staff actually complain about responsiveness.

6.3 Managed hosting, for later

Kept here as the upgrade path, not the starting point. Vercel Pro (~₹1,760) plus a managed Postgres such as Supabase Pro or Neon (~₹1,670–2,200) comes to ≈ ₹3,500–4,000/month, roughly six times the VPS. What that money buys is managed backups, automatic scaling and no server administration.

That trade is not worth making at one shop with ten branches, for three reasons: the load is a few percent of one small server (\$7), serverless adds cold starts to the billing screen and a connection-pool problem to solve, and pg-boss needs a long-lived worker that serverless does not provide. Revisit it when the ops burden genuinely outweighs ₹3,000/month — not before.

6.4 What you are accepting by self-hosting

	Consequence	Mitigation
Backups are yours	Nobody else is protecting the data	Three layers and a monthly restore drill — Deployment Guide §7
Single point of failure	Server dies, app and database go together	Rebuild + restore in 30–60 minutes, practised in advance
OS and patching are yours	Security updates, certificate checks, disk space	unattended-upgrades , Caddy auto-renews, ~2–4 hours a month

The honest cost of self-hosting is not risk, it is **your time**: about half a day to set up and two to four hours a month thereafter. That buys back roughly ₹35,000 a year.

6.5 Portability rules that keep the exit route open

Self-hosting is only the cheap option if leaving stays cheap. Four rules:

1. **Plain PostgreSQL only.** No provider-specific database features. The same schema and migrations must run on your VPS, on Supabase, on Neon or on RDS.
2. **Your own session table**, never a hosting provider's auth service. This is the rule that most often traps people on a platform.
3. **S3-compatible storage behind a small internal interface**, so Cloudflare R2 ↔ any other bucket is a config change.
4. **The app runs in Docker locally.** If it runs in a container on your laptop, it runs anywhere.

Moving to managed hosting later is then `pg_dump` → `pg_restore` → change `DATABASE_URL`. One evening.

6.6 What actually grows the bill

Driver	Effect	Mitigation
Uploaded bill photos and product images	The fastest-growing line, and the one that could push R2 past its free 10 GB	Compress and resize on upload; cap file size; archive attachments older than three years
Database size (device events, ledgers)	~2 M events over five years is only a few GB — modest	Partition <code>device_event</code> by year if it ever matters
Backup retention	Compressed dumps defeat deduplication and multiply storage	<code>pg_dump -Fc -Z 0</code> and let restic compress; 7 daily / 4 weekly / 6 monthly
Long-range analytics queries	CPU, and the temptation to rent a bigger box	The nightly rollups in §4.8 are the fix; buy indexes before hardware
Headless-Chrome PDF generation	Would need a bigger server on its own	Keep to <code>@react-pdf/renderer</code> and browser printing
Adding WhatsApp / SMS notifications	Per-message cost, typically ₹0.20–₹0.90 each	Decide via PRD OQ-6; keep messaging opt-in per rule

At the PRD §9.1 sizing (10 branches, 50 users, ~500 bills a day) none of this needs an upgrade. The first real increase arrives only if attachments outgrow R2's free tier.

7. Performance and Scale

Running the backend inside Next.js on one small server is not a constraint at this size. The arithmetic, taking the PRD sizing and doubling it:

	Estimate
Requests per day (billing, search, dashboards, browsing)	~100,000
Average across a 12-hour trading day	~2.5 req/s
Realistic peak, every branch busy	~30 req/s
What one always-on Node process serves, with a DB query each	200–500 req/s
Database size after five years	a few GB
IMEI lookup on a unique index over 2 M rows	sub-millisecond

Peak load is roughly **5–10% of one CX22**. The shop would need to grow twenty to fifty times before the architecture, rather than a tuning detail, became the question.

7.1 What actually degrades first

1. **Unbounded report queries.** "All branches, all products, three years" scanning everything is the only thing here that genuinely hurts — and it hurts identically in any language or framework. The nightly rollups in §4.8 exist for this.
2. **N+1 queries.** A device list that fetches identifiers per row turns one query into two hundred. Catch it in review; log slow queries in development.
3. **Memory.** A report that materialises a large result set gets the process killed by the OOM killer. Stream exports; paginate everything; never `SELECT *` a whole table into JavaScript.
4. **Disk filling up.** The most common way a small server actually stops working. Old Docker images are the usual culprit — `docker image prune` runs on every deploy.

Note what is *not* on this list: the framework, the language, and the fact that the backend and the frontend share a project.

7.2 Headroom and the upgrade path

Trigger	Response	Downtime
CPU or RAM consistently above 70%	Resize to CX32 (4 vCPU / 8 GB) in the Hetzner console	~1 minute, one reboot
Database outgrows the disk	Attach a Hetzner volume, or resize	Minutes
Sustained load above ~100 req/s	Split the database onto its own server	An evening
Second team, separate deploy cadence	Extract <code>/server/services</code> as its own API	Weeks — and only then

This is why every business rule lives in `/server/services` and never inside a route handler: the split stays cheap without being paid for now.

8. Testing Strategy

Level	Tool	What it covers
Unit	Vitest	Money arithmetic, tax computation, aging buckets, expected-cash calculation, GLOBAL/NEW CUT validation
Service / integration	Vitest + a real Postgres in Docker	Every service function against a real database, including transaction rollback and the concurrent-sale race
Authorisation	Vitest, table-driven	Every endpoint × every role × in-scope and out-of-scope branch
End-to-end	Playwright	The six core business flows in PRD §7, run on every push
Data integrity	Scheduled job	Recompute every ledger balance and compare with what the UI shows; alert on any drift

The concurrency test and the authorisation matrix are the two suites that repay their cost repeatedly. Write them early.

9. Risks and Mitigations

Risk	Impact	Mitigation
ER and ACT are still undefined (PRD OQ-1)	Wrong stock model, discovered after M4 when it is expensive	Answer before M2 starts. Model <code>main_type</code> as an enum + a per-type attribute table so a definition change is a data change, not a migration of every table
The device/event schema is rewritten mid-build	Weeks lost; history gaps that cannot be reconstructed	Design M0–M2 schema fully up front; treat <code>device_event</code> as an API with a version, and never mutate it
Reports built ad hoc per screen	Same number computed three ways, three different answers	One analytics query layer with a shared date/branch/grouping contract (M10)
A device turns out to need a second or third IMEI after launch	Would be a migration plus a rewrite of every form, query and report if identifiers were columns	Already handled: identifiers are rows from the first migration (§4.4) and the UI limit is the <code>imei_slots</code> setting
Backups are never tested	The one failure that turns an outage into a dead business	Monthly restore drill, in M13's "Done when" and in the calendar — Deployment Guide §7.2
Server disk quietly fills up	Postgres stops accepting writes and the shop cannot bill	<code>docker image prune</code> on every deploy; weekly disk check with an alert at 80%
The other system's Excel format changes, or was never as regular as assumed	The daily feed breaks and stock silently drifts	One adapter plus a JSON column mapping; fixture tests built from real exports; the reconciliation report surfaces drift the next morning
The shop stops uploading the daily file	Stock, cash and closing all go wrong quietly	Closing is gated on the feed; a standing dashboard alert while it is missing
Solo-developer bus factor	The project stops	Everything in git, migrations in the repo, <code>README</code> with the runbook, no manual production changes ever. A second SSH key stored offline
Scope creep from "just one more report"	The 29-week plan becomes 50	The module list is the contract; new requests are queued for v1.1
Offline requirement arrives late (PRD OQ-7)	The sales module is substantially rebuilt	Decide before M4. If it is genuinely needed, budget an extra 3–4 weeks and design the sale as a client-generated, idempotent document from the outset

10. Getting Started Checklist

1. `npx create-next-app@latest --typescript --tailwind --app`
2. `npx shadcn@latest init`, then add: `button card input table dialog dropdown-menu form select command sheet tabs toast badge popover calendar`
3. `npm i drizzle-orm postgres drizzle-kit zod react-hook-form @hookform/resolvers @tanstack/react-query @tanstack/react-table recharts pg-boss date-fns`
4. `npm i -D vitest @playwright/test @types/pg eslint prettier`
5. Docker Compose with Postgres 16 for local development

6. GitHub repository, GitHub Actions for typecheck + lint + test + build, deploy on `main`
7. Provision the Hetzner server and Cloudflare R2 bucket, and wire the environment variables — follow the Deployment Guide
8. Write the M0–M2 schema in full before writing the first screen, `device_identifier` and `imei_slots` included
9. Add the money helper (`bigint` paise) and forbid `number` amounts by lint rule
10. Begin Module M0